

1 Define Data Mining and KDD process

Data Mining: The process of finding useful patterns and knowledge from large amounts of data.

Two basic data mining tasks:

- * Classification

- * Clustering

Knowledge Discovery in Databases (KDD):

1. Data Collection

5. Data Transformation

2. Data Cleaning

6. Data Mining

3. Data Integration

7. Pattern Evaluation

4. Data Selection

8. Knowledge Presentation

Role of Data Preprocessing: It removes errors, missing values, and duplicate data, improving the accuracy of data mining results.

2 Data Cleaning

The process of correcting or removing incorrect and incomplete data.

Two data quality issues:

- * Missing values: Replace with mean, median, frequent value.

- * Noisy data: Use binning or smoothing.

Missing values Example:

Age = ? $\rightarrow$ Replace with average age.

Noisy data Example:

20, 21, 80 $\rightarrow$ remove 80 if it is an outlier.

3) Data Integration

Combining data from different sources into one dataset.

Two challenges:

- * Data redundancy: Remove duplicate records.
- * Data inconsistency: Standardize formats and correct conflicting values.

4) Covariance Analysis

Measures how to variables change together.

Two covariance forms:

1. Positive covariance
2. Negative covariance

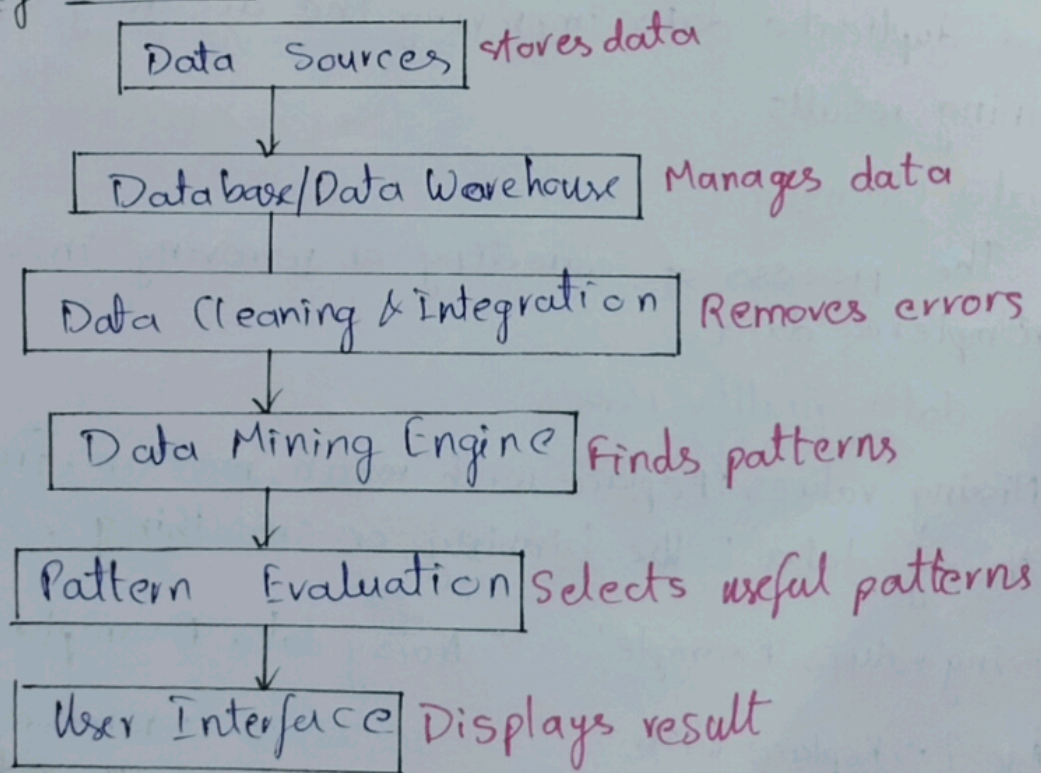
5) Stock Prices

Given $(2, 5), (3, 8), (5, 10), (4, 11), (6, 14)$

Both stock prices generally increase together.

Positive covariance (they rise together, not independently).

6) Data Mining Architecture



7 Covariance

Given,

$$X = \{4, 5, 6, 7, 9\} \quad Y = \{8, 3, 7, 5, 6\}$$

$$\text{Mean}(X) = 6.2$$

$$\text{Mean}(Y) = 5.8$$

$$\text{Covariance} = -0.36$$

Weak negative relationship.

8 Data Preprocessing

Major steps:

1. Data Cleaning
2. Data Integration
3. Data Transformation
4. Data Reduction
5. Data Discretization

Pipeline: Missing values $\rightarrow$ Remove outliers $\rightarrow$ Correct formats
 $\rightarrow$ Normalize data $\rightarrow$ Ready for mining.

Reason: Clean data gives better and more accurate result

9 Equal-Frequency (Equi-Depth) Binning

Sample Data: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25,
30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

Bins (9 values each):

Bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22

Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35

Bin 3: 35, 35, 35, 36, 40, 45, 46, 52, 70

Purpose: Reduces noise while keeping equal number of values in each bin.

10 Data Discretization

Converting continuous data into intervals (bins).

Importance:

- * Reduces data complexity.
- * Improves mining performance.

Equal-Width Binning	Equal-Frequency Binning
Equal interval size	Equal number of values
Simple method	Balanced distribution

Example:

Data: 10, 20, 30, 40, 50, 60

Equal-width: 10-30, 31-60

Equal-frequency: (10, 20, 30) and (40, 50, 60)

Impact:

Equal-width is simple, while equal-frequency handles uneven data distribution better.